

IIT223-2

Information Technology

Project Management

Department of Computer Science and Informatics
Uva Wellassa University

Course Title : Information Technology Project Management

Course Code : IIT223-2

Course Objective : To provide the knowledge of key components of project management and address the characteristics, techniques and problems associated with initiating, planning, executing, controlling and closing IT projects successfully.

Recommended Reading : Kathy Schwalbe, 2018, Information Technology Project Management, 9th Edition

Evaluation Criteria : Continuous Assessments – 40%

End Semester Examination – 60%

Fundamentals of Project Management

CHAPTER 01

Why Project Management ?

- IT projects have become increasingly complex and important, involving; Hardware, Software, Networks to create a product, service, or result.
- Modern organizations now recognize the need to adopt modern project management techniques, especially for IT projects.

Why Project Management ?

- Individuals are realizing that to remain competitive in the workplace, they must develop skills to;
 - become effective project team members and project managers
 - stay competitive in the workplace
 - apply project management concepts in everyday life while working with people and technology

Why Project Management ?

- Organizations using project management techniques report several advantages;
 - Better control of financial, physical, and human resources
 - Improved customer relations
 - Shorter development times
 - Lower costs and improved productivity
 - Higher quality and increased reliability
 - Higher profit margins
 - Better internal coordination
 - Positive impact on meeting strategic goals
 - Higher worker morale

What is a Project ?

Project is “a temporary effort undertaken to create a unique product, service, or result”

(Kathy Schwalbe, 2018, Information Technology Project Management, 9th Edition, p.4)



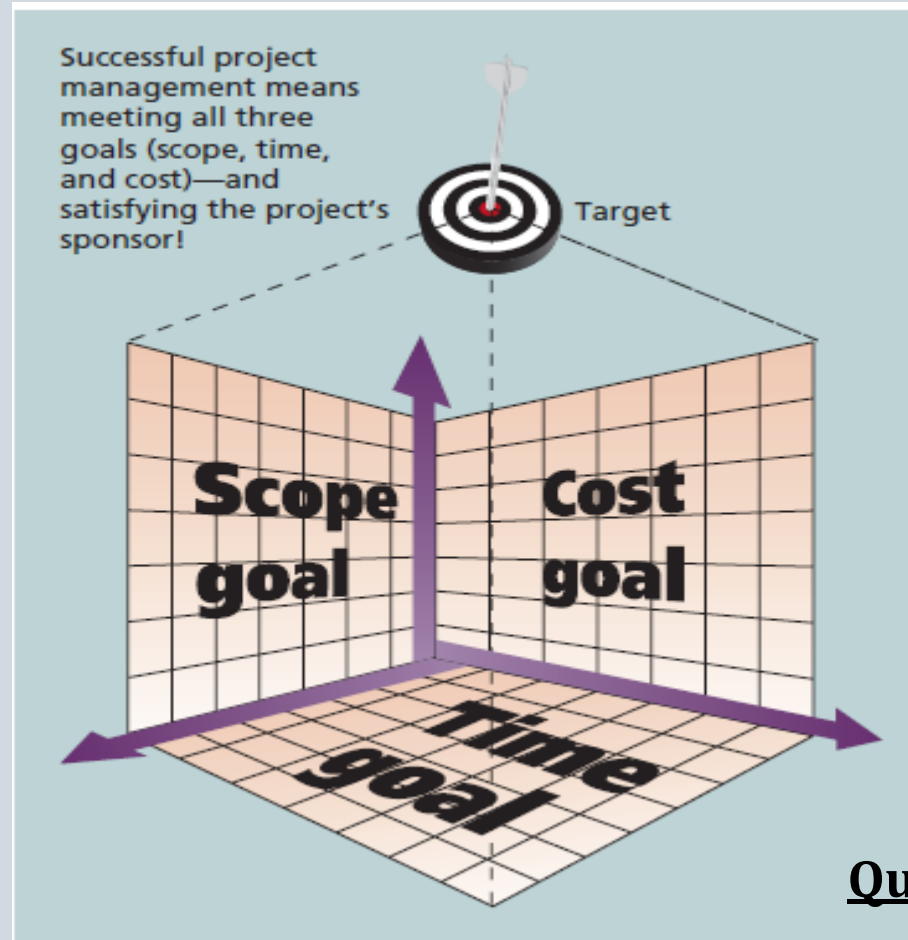
Examples of IT Projects

- A team of students creates a smartphone application and sells it online.
- A company develops a driverless car.
- A university upgrades its technology infrastructure to provide wireless Internet access across the whole campus
- A government group develops a system to track child immunizations.
- Government regulations require monitoring of pollutants in air and water

Project Attributes

- A project has a unique purpose.
- A project is temporary.
- A project drives change and enables value creation.
- A project is developed using progressive elaboration.
- A project requires resources; people, hardware, software, and other assets.
- A project should have a primary customer or sponsor.
- A project involves uncertainty.

The Triple Constraint of Project



Quality is the quadruple constraint

Project Constraints

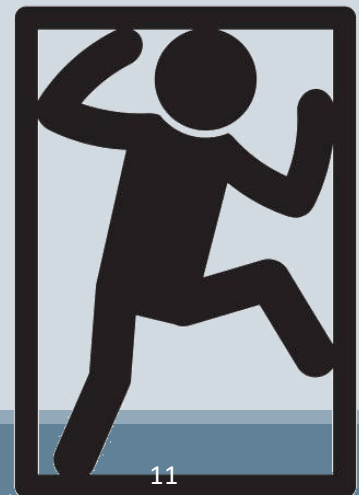
To create a successful project, a project manager must consider scope, time, and cost and balance these three often-competing goals:

- **Scope:**

What work will be done as part of the project?

What unique product, service, or result does the customer or sponsor expect from the project?

How will the scope be verified?



Project Constraints

To create a successful project, a project manager must consider scope, time, and cost and balance these three often-competing goals:

- **Time:**

How long should it take to complete the project?

What is the project's schedule?

How will the team track actual schedule performance?

Who can approve changes to the schedule?



Project Constraints

To create a successful project, a project manager must consider scope, time, and cost and balance these three often-competing goals:

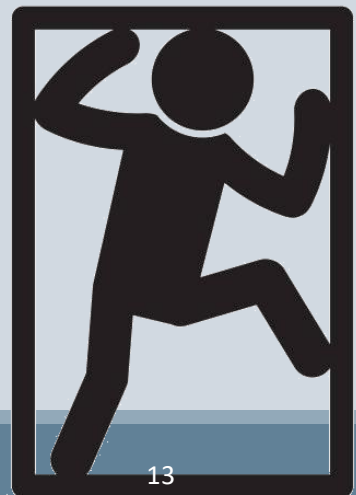
- **Cost**

What should it cost to complete the project?

What is the project's budget?

How will costs be tracked?

Who can authorize changes to the budget?



How can you avoid the problems that occur when you meet scope, time, and cost goals, but lose sight of customer satisfaction?

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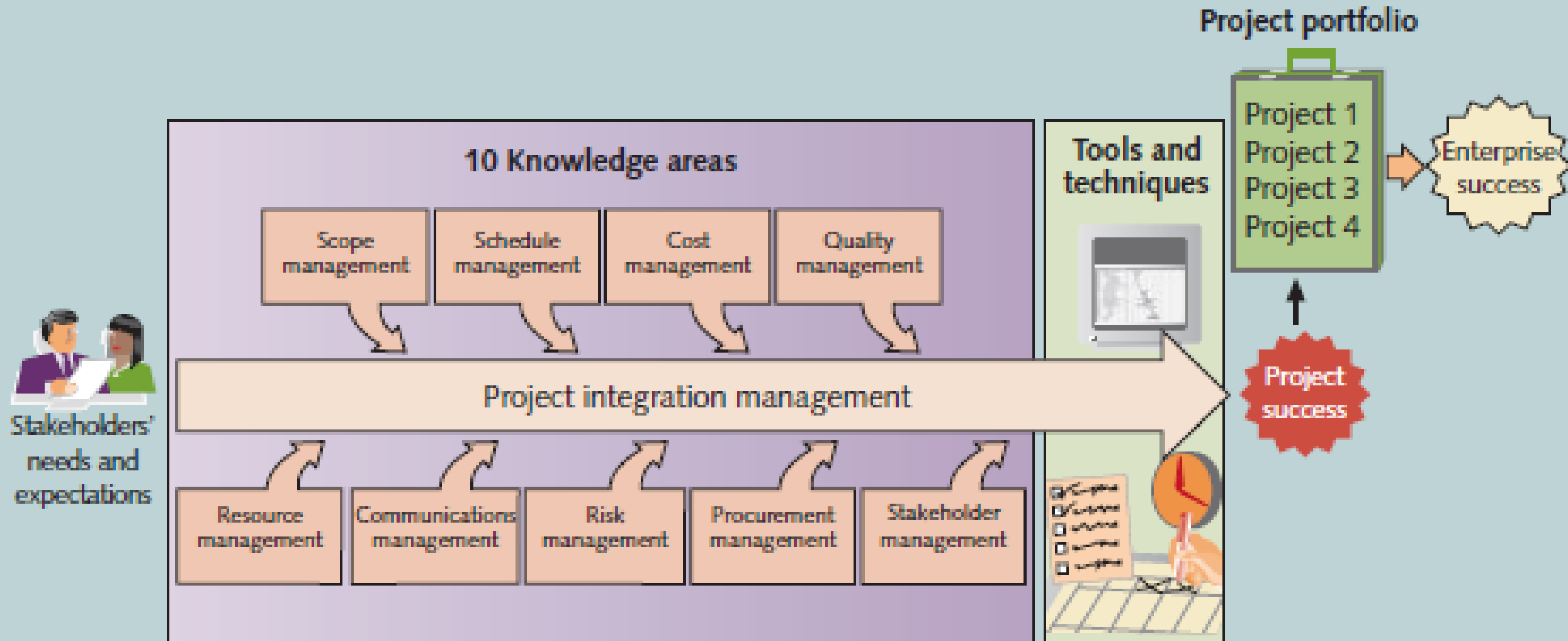
By good project management practices, which includes more than managing project constraints.

What is Project Management?

Project management is “the application of knowledge, skills, tools and techniques to project activities to meet project requirements”

Project managers must strive not only to meet specific scope, time, cost, and quality goals of projects, they must also facilitate the entire process to meet the needs and expectations of people involved in project activities or affected by them.

Project Management Framework



Project Stakeholders

Stakeholders are the people involved in or affected by project activities,

- The project sponsor
- The project manager
- The project team
- Support staff
- Customers
- Users
- Suppliers

Project Management Knowledge Areas

describe the key competencies that project managers must develop

1. **Project scope management** involves defining and managing all the work required to complete the project successfully.
2. **Project schedule management** includes estimating how long it will take to complete the work, developing an acceptable project schedule, and ensuring timely completion of the project.
3. **Project cost management** consists of preparing and managing the budget for the project.
4. **Project quality management** ensures that the project will satisfy the stated or implied needs for which it was undertaken.
5. **Project resource management** is concerned with making effective use of the people and physical resources involved with the project.

Project Management Knowledge Areas

describe the key competencies that project managers must develop

6. **Project communications management** involves generating, collecting, disseminating, and storing project information.
7. **Project risk management** includes identifying, analyzing, and responding to risks related to the project.
8. **Project procurement management** involves acquiring or procuring goods and services for a project from outside the performing organization.
9. **Project stakeholder management** includes identifying and analyzing stakeholder needs while managing and controlling their engagement throughout the life of the project.
10. **Project integration management** is an overarching function that affects and is affected by all of the other knowledge areas.

PM Tools and Techniques

assist project managers and teams to carry out work in all 10 knowledge areas.

Example;

some popular time-management tools and techniques include

Gantt charts

Project network diagrams

Critical path analysis

Knowledge Area/Category	Tools and Techniques	Super Tools	Knowledge Area/Category	Tools and Techniques	Super Tools
<i>Integration management</i>	Project selection methods Project management methodologies Stakeholder analyses Work requests Project charters Project management plans Change control boards Project review meetings	Project management software Change requests Lessons-learned reports	<i>Cost management</i>	Project budgets Net present value Return on investment Payback analysis Earned value management Project portfolio management Cost estimates Cost management plans Cost baselines	
<i>Scope management</i>	Statements of work Scope management plans Scope verification techniques Scope change controls	Scope statements Work breakdown structures Requirements analyses	<i>Quality management</i>	Quality metrics Checklists Quality control charts Pareto diagrams Fishbone diagrams Maturity models Statistical methods Test plans	
<i>Schedule management</i>	Project network diagrams Critical path analysis Crashing Fast tracking Schedule performance measurements	Gantt charts			

Knowledge Area/Category	Tools and Techniques	Super Tools
<i>Resource management</i>	Motivation techniques Empathic listening Responsibility assignment matrices Project organizational charts Resource histograms Team building exercises	
<i>Communications management</i>	Communications management plans Conflict management Communications media selection Status reports Virtual communications Templates Project websites	Kick-off meetings Progress reports
<i>Risk management</i>	Risk management plans Risk registers Probability/impact matrices Risk rankings	
<i>Procurement management</i>	Make-or-buy analyses Contracts Requests for proposals or quotes Source selections Supplier evaluation matrices	

Project Success

There are several ways to define project success

- The project met scope, time, and cost goals
- The project satisfied the customer/sponsor even if the scope, time and/or costs goals were not met
- The results of the project met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy.

What Helps Projects Succeed?

1. Executive support
2. User involvement
3. Experienced project manager
4. Clear objectives
5. Minimized scope
6. Standard software infrastructure
7. Formal methodology
8. Reliable estimates
9. Other criteria, such as small milestones, proper planning, competent staff, and ownership

Project Portfolio Management

As part of **project portfolio management**, organizations group and manage projects and programs as a portfolio of investments that contribute to the entire enterprise's success

Portfolio managers help their organizations make wise investment decisions by helping to select and analyze projects from a strategic perspective

Project Management Compared to Project Portfolio Management



Project Manager

- Project managers are organized, passionate, and goal-oriented individuals who drive business results by leading projects.
- they are also change agents who work well under pressure and enjoy challenging work environments.
- Project management jobs can be found in every country and every industry

Suggested Skills for Project Managers

Project managers need a wide variety of skills such as;

- Be comfortable with change
- Understand the social, political and physical environments of the organizations they work in
- Be able to lead teams to accomplish project goals
- Possess general management knowledge and skills
- Soft skills or human relations skills
- Teamwork skills
- Strong leadership skills

Suggested Skills for Project Managers

Project managers need a wide variety of skills such as;

- flexible, creative, and sometimes patient in working toward project goals
- able to make effective use of technology as it relates to the specific project
- make many decisions and deal with people in a wide variety of disciplines

The end.
